Hardik Gupta

hg08072005@gmail.com +91 9717417305 hg-8 Hardik Gupta Bengaluru, Karnataka

EDUCATION

PES University, *B.Tech in Electronic and Communication Engineering* Aug 2023 – Present — Bangalore, India

Lotus Valley International School, *Class 12th- CBSE* Apr 2022 – Mar 2023 — Noida, India

Lotus Valley International School, *Class 10th- CBSE* Apr 2020 – Mar 2021 — Noida, India

CERTIFICATION

AI & ML with Drone Technology - IIT Hyderabad (Tihan) Jun 2025 – Feb 2026

- Hands-on training in **ML, Deep Learning,** and AI deployment with academic exposure from **IIT Hyderabad.**
- Built and evaluated **regression, classification, clustering,** and **neural network** models using Python libraries.
- Gained practical understanding of the building blocks and working principles of drones, including core components.

EXPERIENCE

Bharat Space Education Research Centre, New Delhi Dec 2025 – Jan 2026

Winter Intern

- Modeled aircraft stability and performance using **MATLAB (AID toolbox) and XFLR5**, identifying key design parameters for optimal flight behavior.

ColorHome Lighting, Noida Jul 2025 – Aug 2025

Summer Intern

- Analyzed the end-to-end manufacturing lifecycle of 9W LED bulbs, from component sourcing to final dispatch.
- Gained hands-on experience in **SMT operations, quality assurance, and electrical safety testing.**

PROJECTS

Vocab Ledger – Native Android Vocabulary App Jul 2026

- Architected an offline-first native Android app in **Kotlin & Compose** with **Google Gemini AI API** integration for real-time mnemonics and smart dictionary fallback; published on **Google Play Store**.
- Optimized release binary size by **44%** (from 19.7 MB down to 11.0 MB) by configuring custom **R8/ProGuard** minification and resource shrinking rules.

Automated PCB Defect Detection System Apr 2026

- Trained and benchmarked **YOLOv5s** and **YOLOv8s** on 693 PCB images to detect and classify 6 manufacturing defect types, achieving **97.4% mAP@0.5** and **95.2% recall**.
- Built a 4-level severity analysis system that evaluates defect area, position, and confidence to deliver automated per-board PASS/FAIL verdicts with repair recommendations.

Automatic Audio Synthesis for Silent Videos Mar 2026

- Implemented the AutoFoley Frame-Relation Network (**ResNet-50 + Multiscale TRN**) to predict and synthesize audio for silent videos across 10, 12, and 15 sound-class configurations using VGGSound and AVSync15 datasets.
- Built an end-to-end pipeline covering frame extraction, per-class average spectrogram computation, temporal relation-based classification, and **Griffin-Lim audio reconstruction** from predicted spectrograms.

Federated AI for Privacy-Preserving Alzheimer's Detection Aug 2025 – Present

- **Designing a secure Federated Learning pipeline** for Alzheimer's MRI classification using **VGG19/ResNet50**, enabling decentralized model training on private data.
- **Integrating Grad-CAM and SHAP** to generate visual heatmaps and feature importance scores, improving the interpretability of the model's predictions.

SKILLS

Languages & Mobile: Python, C, Kotlin, Assembly, Jetpack Compose, Room DB, Firebase

Machine Learning & CV: Scikit-learn, TensorFlow, PyTorch, OpenCV, YOLOv5/v8, NumPy, Pandas, Matplotlib, Seaborn

Tools & Frameworks: Git, Jupyter, VS Code, MATLAB, Google Colab, Ripes, Arduino IDE, LaTeX